

LAB 1

1. Write a Java program to print "Hello, World!" to the console.

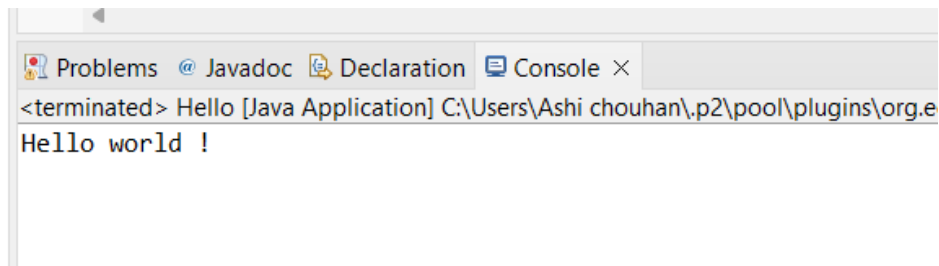
PROGRAM:

```
package nDemo;

public class Hello {

    public static void main(String[] args) {
        // Printing "Hello world" message in the console
        System.out.println("Hello world !");
    }
}
```

OUTPUT:



2. Write a program to find the sum of two numbers entered by the user.

PROGRAM:

```
package nDemo;
import java.util.Scanner;

public class Sum {

    public static void main(String[] args) {
        // Scanner object
        Scanner sc = new Scanner(System.in);

        System.out.println("Enter int a : ");
        // Input from user of integer a
        int a = sc.nextInt();

        // Input from user of integer b
        System.out.println("Enter int b : ");
        int b = sc.nextInt();

        sc.close();

        // Calculating sum of a and b
    }
}
```

```

        int result = a + b;
        // Printing result
        System.out.println("The sum of a and b is : " + result);
    }
}

```

OUTPUT:

```

<terminated> Sum [Java Application] C:\Users\Ashi chouhan\p2\pool\plugin
Enter int a :
10
Enter int b :
20
The sum of a and b is : 30

```

3. Write a Java program to check whether a given number is even or odd.

PROGRAM:

```

package NDemo;
import java.util.Scanner;

public class EvenOdd {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.println("Enter int a : ");
        double a = sc.nextDouble();

        if( a % 2 == 0 ) {
            System.out.println("The given number a is even.");
        }
        else {
            System.out.println("The given number a is odd.");
        }
    }
}

```

OUTPUT:

```

<terminated> EvenOdd [Java Application] C:\Users\Ashi chouhan\p
Enter int a :
31
The given number a is odd.

```

4. Write a java program to find greatest of 3 numbers.

PROGRAM:

```
package nDemo;
import java.util.Scanner;

public class greatestOfThreeNumber {

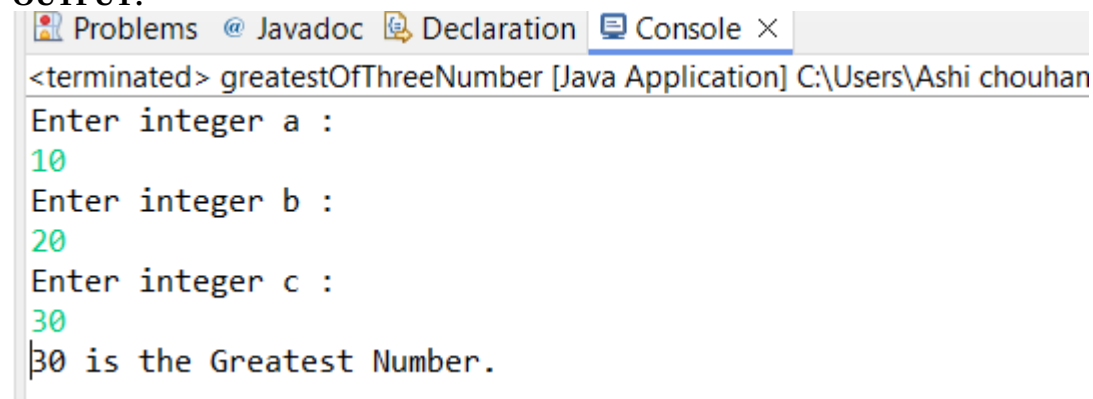
    public static void main(String[] args) {

        // Scanner object to take input
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter integer a : ");
        int a = sc.nextInt();
        System.out.println("Enter integer b : ");
        int b = sc.nextInt();
        System.out.println("Enter integer c : ");
        int c = sc.nextInt();

        // closing scanner object
        sc.close();

        // If condition to check the greatest number
        if(a > b && a > c) {
            System.out.println(a + " is the Greatest Number. ");
        } else if (b > a && b > c) {
            System.out.println(b + " is the Greatest Number. ");
        } else {
            System.out.println(c + " is the Greatest Number. ");
        }
    }
}
```

OUTPUT:



```
<terminated> greatestOfThreeNumber [Java Application] C:\Users\Ashi chouhan
Enter integer a :
10
Enter integer b :
20
Enter integer c :
30
30 is the Greatest Number.
```

5. Write a program to implement a basic calculator that takes input and evaluates it.

PROGRAM:

```
package nDemo;

// Class containing calculator methods
class calculatorMethods {

    // Add method
    public int add( int x, int y) {
        return x + y;
    }

    // Subtraction method
    public int sub( int x, int y) {
        return x - y;
    }

    // Division method
    public int div( int x, int y) {
        return x / y;
    }

    // Multiplication method
    public int product( int x, int y) {
        return x * y;
    }
}

public class Calculator {

    public static void main(String[] args) {

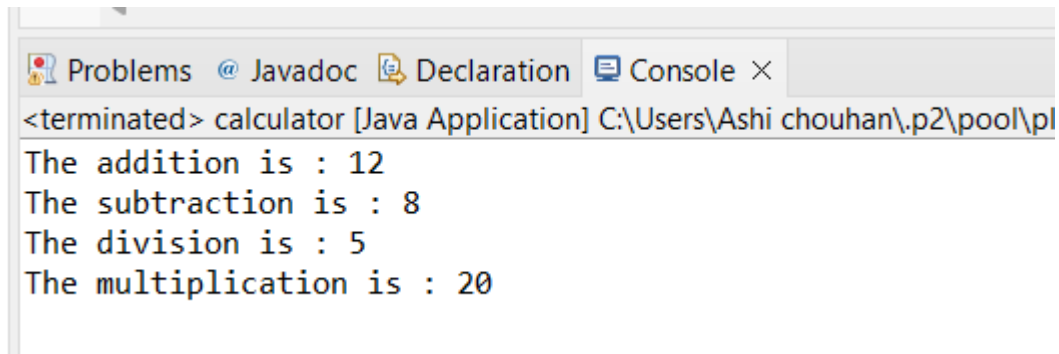
        // Taking 2 integer values a and b
        int a = 10;
        int b = 2;

        // creating object of class "calculatorMethods"
        calculatorMethods cm = new calculatorMethods();

        // Printing evaluated values
        System.out.println("The addition is : " + cm.add(a, b));
        System.out.println("The subtraction is : " + cm.sub(a, b));
        System.out.println("The division is : " + cm.div(a, b));
        System.out.println("The multiplication is : " + cm.product(a, b));

    }
}
```

OUTPUT:



```
<terminated> calculator [Java Application] C:\Users\Ashi chouhan\.p2\pool\pl
The addition is : 12
The subtraction is : 8
The division is : 5
The multiplication is : 20
```

6. Write a Java program to check if a given number is prime or not.

PROGRAM:

package nDemo;

import java.util.Scanner;

public class PrimeNumberChecker {

public static void main(String[] args) {

 // Creating a Scanner object for user input

 Scanner scanner = **new** Scanner(System.**in**);

 System.**out**.print("Enter a number to check if it is prime: ");

int number = scanner.nextInt();

 // Closing the scanner

 scanner.close();

 // Initiating boolean value for prime number as true

boolean isPrime = **true**;

 // Method to check if entered number is prime or not

if (number <= 1) {

 isPrime = **false**;

 } **else** {

for (**int** i = 2; i < number; i++) {

if (number % i == 0) {

 isPrime = **false**;

break;

 }

 }

 }

 // Displaying the result

if (isPrime) {

 System.**out**.println(number + " is a prime number.");

 } **else** {

 System.**out**.println(number + " is not a prime number.");

 }

 }

}

OUTPUT:

```
<terminated> PrimeNumberChecker [Java Application] C:\Users\Ashi
Enter a number to check if it is prime: 12
12 is not a prime number.
```

7. Create a Java program that compares two numbers and prints the larger one.

PROGRAM:

```
package nDemo;
import java.util.Scanner;

public class GreaterNumber {

    public static void main(String[] args) {
        // Scanner object
        Scanner sc = new Scanner(System.in);

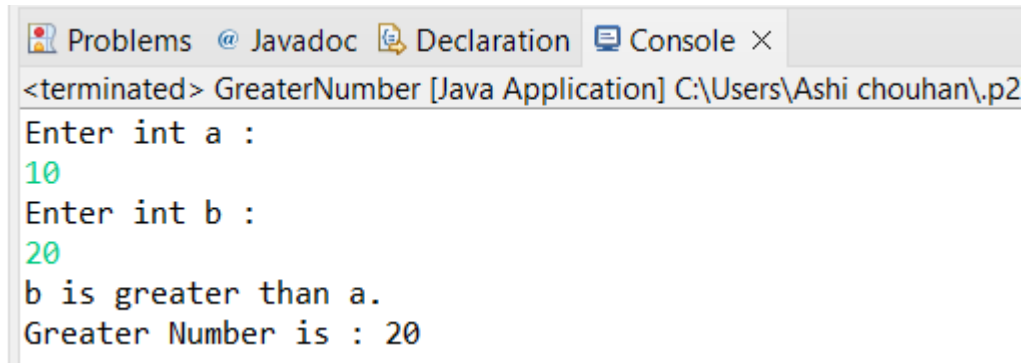
        // Taking integer input for a and b
        System.out.println("Enter int a : ");
        int a = sc.nextInt();

        System.out.println("Enter int b : ");
        int b = sc.nextInt();

        // Closing scanner object
        sc.close();

        // Declaring result variable
        int result;
        if ( a > b ) {
            result = a;
            System.out.println("a is greater than b.");
            System.out.println("Greater Number is : " + result);
        }
        else if ( b > a ){
            result = b;
            System.out.println("b is greater than a.");
            System.out.println("Greater Number is : " + result);
        }
        else {
            System.out.println("a is equal to b.");
        }
    }
}
```

OUTPUT:



```
<terminated> GreaterNumber [Java Application] C:\Users\Ashi chouhan\.p2
Enter int a :
10
Enter int b :
20
b is greater than a.
Greater Number is : 20
```

8. Write a Java program that takes an age input from the user and determines if they are eligible to vote (considering the legal voting age).

PROGRAM:

```
package nDemo;
import java.util.Scanner;

// Eligibility checker class
class votingEligibility {

    public void eligibility(int age) {

        if (age < 100 ) {
            if ( age < 18 ) {
                System.out.println("Not Eligible !");
            }
            else if ( age >= 18 ) {
                System.out.println("Eligible !");
            }
        }
        else {
            System.out.println("Invalid age !");
        }
    }
}

public class VotingAge {

    public static void main(String[] args) {

        // Scanner object
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter your age to check if you are eligible to vote : ");
        // Age input
        int age = sc.nextInt();
        sc.close();
    }
}
```

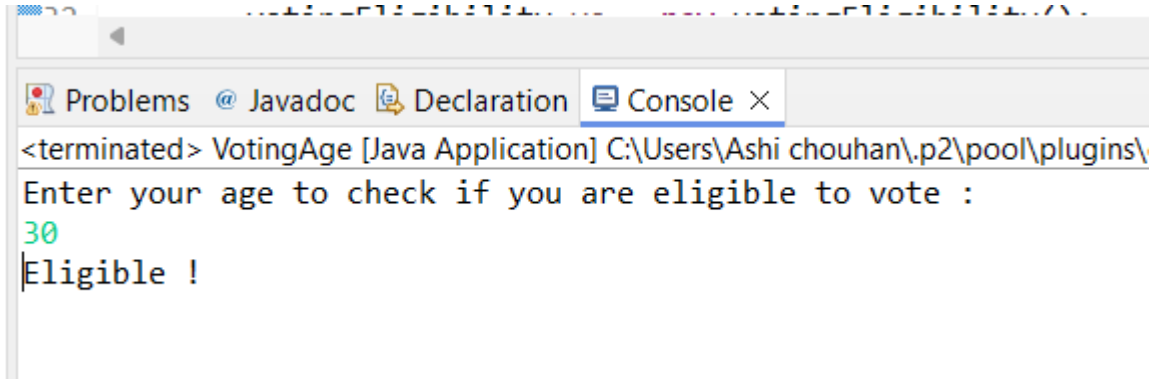
```
// "votingElibility" class object
votingElibility ve = new votingElibility();

// Calling eligibility() function from "votingElibility" class using object.
ve.eligibility(age);

    }

}
```

OUTPUT:



The screenshot shows an IDE console window with the following content:

```
<terminated> VotingAge [Java Application] C:\Users\Ashi chouhan\.p2\pool\plugins\
Enter your age to check if you are eligible to vote :
30
Eligible !
```